AccuKnox Assignment-1

Task-1

Technologies Used :Python,SQLite

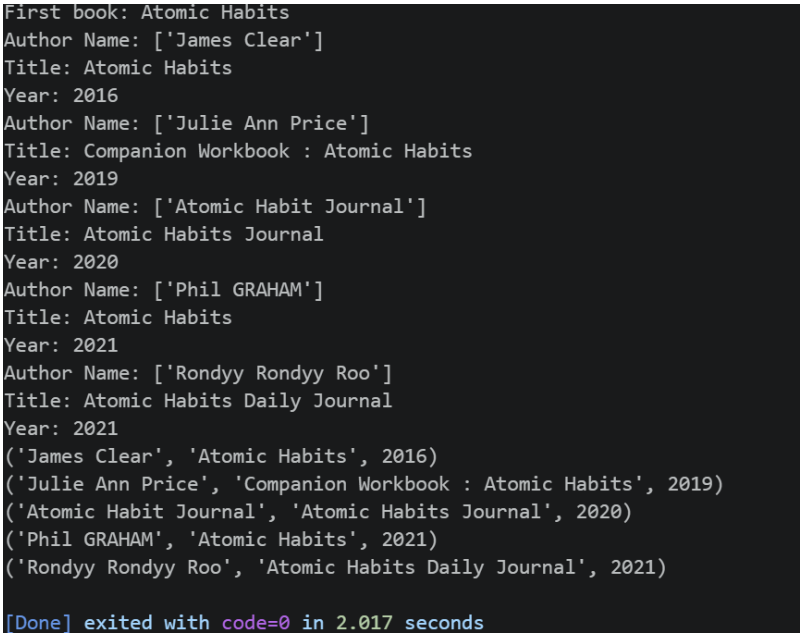
Objective:

Retrieve book information from an external API, process the JSON response, store selected book details in a local SQLite database, and display the stored records.

Approach:

1)I used Python's requests library to send a GET request to the Open Library API.
2)I converted the API response into JSON format using the response's .json() method.
3)I extracted book titles, author names, and publication years from the response.
4)I used Python's sqlite3 module to create a local database and a table for storing the book information.
5)I inserted the first five book records using Cursor for executing code in Python File.
6)I retrieved the stored records using a SELECT query and displayed them in the terminal

Output: The program prints the API response information, selected book details, and records retrieved from SQLite.



Task-2

Technologies Used :Python,Matplotlib,Pandas

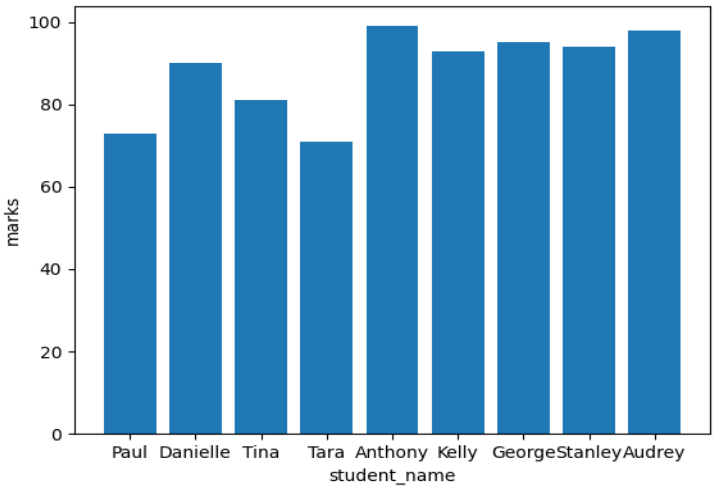
Objective:

To retrieve student test scores from an external API, calculate the average mathematics score, and visualize the scores of students using a bar chart.

Approach:

1) I used Python's requests library to send a GET request to the Sling Academy student scores API.
2) I converted the API response into JSON format and inspected the returned data to understand its structure.
3) I created two empty lists all_marks and students_name .Later, I iterated through the retrieved records, extracting each student's mathematics score and first name using the . get() method and storing them in the respective lists.
4) I calculated the total of all mathematics scores using Python's sum() function. I divided this total by the number of records to calculate the average mathematics score.
5) I used Matplotlib to create a bar chart showing the mathematics scores of the first ten students.

Output: A bar chart representing the mathematics scores of the first ten students.



Task-3

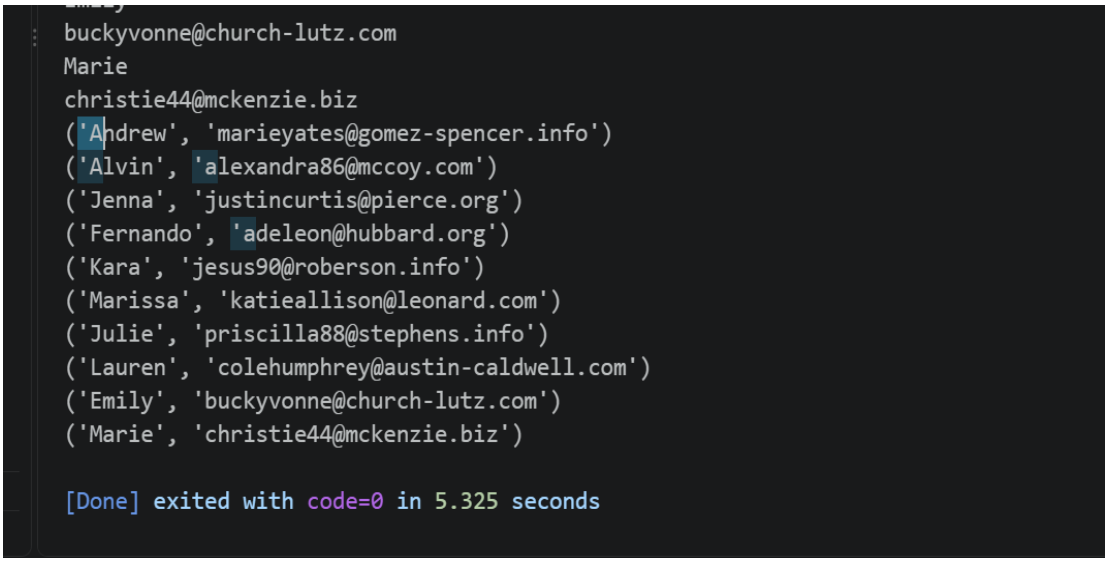
Technologies Used :Python,SQLite

Objective: To read customer information from a CSV file, extract names and email addresses, store the selected records in a local SQLite database, and retrieve the stored information.

Approach:

1) I used the Pandas library to read a csv file into a DataFrame.
2) I selected the First Name and Email columns from the DataFrame because these were the fields required for the task.
3) I used Python's sqlite3 module to connect to a database and created a table called info with two columns Name and Email.
4) I iterated through the first ten rows of the DataFrame using iterrows(). For each row, I extracted the customer's first name and email address and inserted them into the info table using a SQL query.
5) After inserting the records I used commit() to save the database changes.
6) I executed a SELECT query to retrieve the names and email addresses from the info table and displayed the results using a loop.

Output: The program displays names and email addresses extracted from the first ten rows the records are retrieved from the SQLite database.



Task-4 :Most Complex Code

Project**:** AdvocadabraLLM

Objective:

- The objective of this code is to develop a machine learning-based Legal Judgment Prediction (LJP) system that analyzes legal case representations and predicts potential case outcomes.

- The module explores how machine learning and similarity search can be combined to classify legal cases into outcome categories such as plaintiff, defendant, and dismissal. It also generates supporting information from similar cases to make the prediction process more interpretable.

- It is a feature Present In Our AdvocadabraLLm application

Source Code: https://github.com/Pluto01/AdvocadabraLLM

Task-5 :Most Complex Database Code

Project**:** Intelligent Support Ticket Router

Overview: This module is part of a Support Ticket Router application. It stores masked support tickets and their classification details, such as category, subcategory, sentiment, priority score, and target tier. Original PII values are stored separately in an encrypted vault.

Source Code: https://github.com/Jye-Prahaas/Intelligent-Support--Ticket-router